

Working Memory Training for Older Participants: A Control Group Training Regimen and Initial Intellectual Functioning Assessment

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Abstract

The efficacy of cognitive training interventions is recently highly debated. There is no consensus on what kind of training regimen is the most effective. Also, individual characteristics as predictors of training outcome are still being investigated. In this article, we show the attempt to address this issue by examining not only the impact of working memory (WM) training on cognitive effectiveness in older adults but also the influence of the initial WM capacity (WMC) on the training's outcome. We describe in detail how to perform 5 weeks of an adaptive dual n-back training with an active control group (memory quiz). We are focusing here on technical aspects of the training as well as on the initial assessment of participants' WMC. The evaluation of pre and post training performance of other cognitive dimensions was based on the results of tests of memory updating, inhibition, attention shifting, short-term memory (STM) and reasoning. We have found that the initial level of WMC predicts the efficiency of the n-back training intervention. We have also noticed the post training improvement in almost all aspects of cognitive functioning we measured, but those effects were mostly intervention independent.

Introduction

In many cognitive trainings studies, the dual n-back task is used as a method of working memory (WM) training. WM is a common target of cognitive interventions because of its importance for other, higher level intellectual functions¹. However, the effectiveness of such training and its potential for creating a more general improvement in cognition, has been highly debated (for meta-analysis, see^{2, 3, 4, 5, 6, 7, 14} and for reviews, see^{4, 8, 9, 10, 11, 12, 13}). While some

researchers claim that "... there was no convincing evidence of the generalization of working memory training to other skills"⁴, others present meta-analytical data, which show highly significant effects of WM training^{2, 3, 5, 6, 11}. The separate problem is the effectiveness of WM in the elderly population. Several WM training studies reported greater benefits in younger adults compared to older

adults^{15, 16, 17, 18, 19, 20}, whereas others show that similar effects can be observed in both age groups^{21, 22, 23, 24, 25}.

Various elements are believed to forecast the benefits of memory training²⁶. Some of those factors appears to be potential moderators of WM training effectiveness²¹. Mental capacity, being described as the baseline cognitive capacity or general cognitive resource, seems to be one of the strongest choices for this position. In order to assess the role of the initial intellectual level, we put a special emphasis (the method described here) on the measurement of the cognitive capacity before applying a training regimen. It was dictated by the data showing that participants, who are characterized by higher cognitive capacity at the beginning of the training, achieved substantially better training outcomes compared to those with lower levels of initial cognitive functioning²⁷. A similar phenomenon is observed in educational research, where it is referred to as the Matthew effect²⁸, an observation that people with initially better skill improve even more when compared to those with preliminary lower level of ability in question.

It is thought-provoking, though, that not so many reports have been published on this topic^{21, 29}. Moreover, even substantial individual differences, especially when it comes to the elderly population, are often left unattended during data analysis and interpretation³⁰. In the present study, we examine the impact of the initial level of working memory capacity on WM training success in the group of healthy older adults. In order to maintain every element of the training regimens as similar as possible between experimental and control groups, we employed an active control group design. Therefore, the training content (WM versus semantic memory) remained the one crucial factor determining the expected difference in the training results.

Both groups performed computerized, home-based trainings. Members of the experimental group were assigned to an adaptive dual n-back training program and an active control group trained with a task based on a semantic memory quiz. New in the approach here is the emphasis on the initial evaluation of the participants' cognitive level by assessing their working memory capacity (WMC). Additionally, the method of assessing the initial WMC level we present in this article has proven to be an effective tool in distinguishing between people who will and will not be successful during subsequent working memory training. We have previously described and published results from this study⁴⁴. Therefore, in this article we are focusing on a detailed description of the protocol we used.

Protocol

The SWPS University of Social Sciences and Humanities Ethics Committee assessed the protocol described here. A written informed consent in accordance with the Declaration of Helsinki was obtained from every participant.

1. Participants recruitment

1. Recruit at least 36 volunteers for each training group. This number was evidenced to be enough to observe between-groups effects by authors' previous research and also in the literature on the subject. Typical effect sizes in working memory trainings studies range between $d=0.6$ to 0.8 depending on the training type or targeted group. Based on these values and aiming at a decent statistical power of 0.8 with alpha at 0.05 , a minimum sample size in this type of research (calculated according to a formula proposed by Soper⁴⁵) is 36 (preferably more).

1. Use the following inclusion criteria: over 55 years old, with no history of neurological or psychiatric disorders, with preserved motor skills of the upper limbs, without blindness or hearing loss, who are not presently involved in any other cognitive (especially memory) training.
2. Recruit participants using various methods: online advertisements posted on social media profiles, research and work platforms or discussion groups, as well as in-person announcements at Universities of a Third Age or during events involving older audience such as picnics for seniors (also with usage of posters and leaflets) to get certainty that the people who are recruited are not only internet users.
3. Remember to adequately describe in the advertisement what type of participants you are looking for.
2. Be sure that participant does not take drugs or has never suffered from any disease that may affect central nervous system functioning. Likewise, control intake of medications not related to neurological diseases that affect cognitive functioning. If the screening revealed any unwanted information, exclude the volunteer from the study.
3. After successful screening, present the written informed consent form to the participant and ask them to read it. The written informed consent should involve information as follows: a) the legal bases for data gathering and processing specific for a given country, b) information about rights of the data owner (e.g., accessing the personal data, possibility to supplement incomplete personal data, deleting data or processing restrictions).
 1. Ask the participant to sign the informed consent.

2. The evaluation of the Ethics Committee

1. Before starting the study, obtain the evaluation form your local Ethics Committee, including permission to: a) Interactions consisting of active intervention in human behavior, aimed at changing this behavior, without directly interfering with the brain, e.g. cognitive training, psychotherapy, etc. (this also applies to intervention intended to benefit the respondent, e.g. improving his memory), b) collect and process participants' personal data, in particular data that allows to identify the subjects.

3. Initial screening

1. Begin with a short interview informing the participant in detail about the project's aim, the possibility of withdrawal, and personal data protection.
4. Carry out the Mini Mental State Examination (MMSE)³² to ensure that participant shows no signs of mild cognitive impairment - at least 27 points are needed to enter next stages of a study.
 1. Read the MMSE's Introductory Script to the participant and then ask questions in accordance with the examination script.
 2. Begin with an assessment of orientation to place and time by asking series of questions: What is today's full date? What day of week is today? Where are we (what city, name the building, which floor)?
 3. Follow with memory test: ask participant to memorize three objects that was read at loud by you; go through the series of seven tasks assessing attention, concentration, and calculation, and at the end ask participant to recall the three objects previously learned.

4. Finally, test naming, repetition, and understanding according to examination script.
 5. Score responses as follows: 0 = incorrect or lack of an answer, 1 = correct answer.
 6. Do not take longer than 10 min for the administration of the MMSE test.
 7. If a person does not reach the required threshold (27 points), inform them of the result. If there is any suspicion of a clinically lowered level of cognitive functioning, refer such a person to a specialist unit (e.g., a certified psychologist at a neurological center).
5. Store documentation in a manner that complies with the law and/or The General Data Protection Regulation of the local country.
- #### 4. Training group assignment
1. Randomly assign participants to an experimental or control group (**Figure 1**). To ensure the randomness of the process, generate a list of 50 codenames (**Figure 2**) assigned to ones and twos (training groups) and connect each participant in recruiting order with those codes (saved in separate file). From now on, replace participant data with codenames.

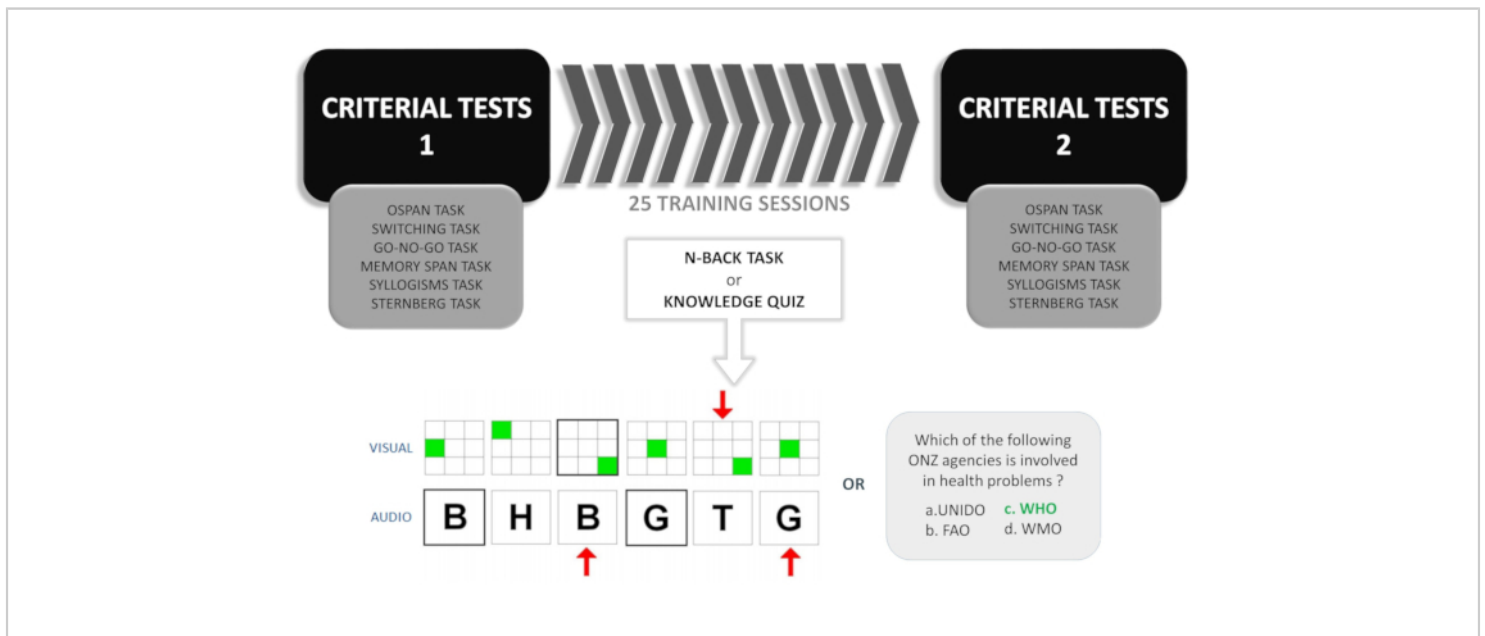


Figure 1. Study design with examples of a training tasks. Participants underwent two measurement sessions, before and after a 5 week training protocol. [Please click here to view a larger version of this figure.](#)

2. Make sure that the group assignment is not biased in terms of age, gender or the level of education. Pre-assign age, gender and education group to the list of code-
- names for each training group (1 and 2), as presented in **Figure 2**. Fit each volunteer in the table based on their characteristics.

code	training	gender	age	education
name1	1	1	1	1
name2	1	1	1	2
name3	1	1	1	3
name4	1	1	2	1
name5	1	1	2	2
name6	1	1	2	3
name7	1	1	3	1
name8	1	1	3	2
name9	1	1	3	3
name10	1	2	1	1
name11	1	2	1	2
name12	1	2	1	3
name13	1	2	2	1
name14	1	2	2	2
name15	1	2	2	3
name16	1	2	3	1
name17	1	2	3	2
name18	1	2	3	3
name19	2	1	1	1
name20	2	1	1	2
name21	2	1	1	3
name22	2	1	2	1
name23	2	1	2	2
name24	2	1	2	3
name25	2	1	3	1
name26	2	1	3	2
name27	2	1	3	3
name28	2	2	1	1
name29	2	2	1	2
name30	2	2	1	3
name31	2	2	2	1
name32	2	2	2	2
name33	2	2	2	3
name34	2	2	3	1
name35	2	2	3	2
name36	2	2	3	3
etc.	...	...	...	...

Figure 2. The example of suggested coding form for group assignment.

5. Initial assessment of cognitive functioning

1. Put an emphasis on giving very clear and detailed instruction to the participants about how to go through each task at the beginning of each procedure. Run the "training block" before each task (that is identical to the

training task) and observe the participant if their answers indicate that they understand the instructions.

NOTE: The inclusion of such blocks is described at the beginning of description of each task below.

2. After presenting the instruction and practice part of each task and before starting the main part of the

procedure, once again ask if the participant understands the procedure's requirements.

3. Be sure that each participant will accomplish the full set of the following tasks.

4. The operation span task (OSPAN)

1. Run a training block to estimate the individual time needed for each participant to calculate a simple mathematical equation (adding, subtraction, dividing and multiplication of single digits - not higher numbers).
2. In the middle of the white screen display the equation to the participant. Instruct the participant to think of the result and then press an arrow leading to the next screen, where the outcome of equation is presented. Let the participant give an answer by pressing either the **True** or **False** button.
3. Count the time needed for resolving the equation. Use mean time needed in a final block as a time of displaying the equation in the main part of the task. Have a fixed time limit imposed for estimating the correctness of the equation result: 5 s.
4. In the next training block display letters on the screen - one by one, for 500 ms each, and instruct the participant to remember them. After a full set (3 to 9 letters), present the matrix of 12 letters to participant and ask to mark memorized letters in a correct order. Do not give a time limit for the answer. Record the correctness.
5. Run a main part of the task. In the final block, mix two abovementioned training blocks: after each part with equation (remember about time limits!) present one letter to remember. Display 2 to 5 pairs of 'equation + letter' and then, after presenting the whole sequence

of 'equation + letter' pairs, show the matrix of letters for participant to mark memorized letters in a correct order. Record the correctness of mathematical and memory part.

NOTE: With this test the operational span of Working Memory (processing the one kind of information while remembering the other) is assessed.

5. The Sternberg task

1. Present a random sequence of digits (2 to 5 in a one set) in white font on black screen, for 500 ms each, with the interval ranging from 2500 to 3000 ms.
2. Display a fixation cross for 2500 ms.
3. At the end of presented sequence, display a target digit in a yellow font for 500 ms.
4. Let the participant decide if the yellow digit appeared among the presented earlier set of white digits by pressing **Yes/No** buttons. If the participant does not give an answer within 3000 ms, go to the screen with fixation point and start the next trial. Count this attempt as a wrong answer.
5. Repeat steps 5.5.1 to 5.5.4 120 times (trials) with 50% of probes containing target digit in the sequence and 50% do not (randomly).
6. Record the correctness and the reaction time for each trial.

NOTE: This task tests the speed of searching the information in memory. The increase of reaction time accompanies the enlargement of the set, which is explained as the process of a serial search of the memory content.

6. The running memory span task

1. On the screen, present information about the number of letters to remember (3, 4, 5 or 6 letters depending on the difficulty level of a block) and ask the participant to go to the next screen by pressing a key.
2. Present a sequence of letters, one by one, in black font in the center of a white screen, for 0.25 s each.
3. Ask the participant to reproduce a fixed number of the last letters from the sequence: fixed no: 4; sequence: K U J D S T W A; letters to memorize: S T W A.
4. To receive the participant's answer, display on the screen a matrix of 9 letters (3x3) and ask the participant to mark appropriate letters (in order the letters appeared) with the mouse. Do not give a time limit for the answer.
5. Record correctness (mind the sequence errors).

NOTE: This test measures the capacity of the working memory with the use of the additional distraction in the form of the inability to predict which letters from the list would be the portion to remember.

7. Test Go-No Go

1. On the white screen display trials composed of: a) 250 ms - a fixation point (white cross), b) 1250 ms - the stimuli (a letter), c) 2000 ms - a fixed inter-stimulus interval.
 2. Have the participant react by pressing a key as quickly as possible, when a target stimulus - letter X - appears on the screen.
 3. Record the reaction time and correctness of answers.
- NOTE:** The test measures the efficiency of inhibition in two conditions: in easier condition the letter X is presented in the 50/50 proportion to other letters and

in the more difficult condition the target stimulus is displayed in 70/30 proportion to other letters.

8. The switching task

1. Divide the screen into two parts using a horizontal line. Present red squares or rectangles composed of smaller squares or rectangles above or below this line.
2. Apply two different rules for the participant to react, depending on where figures will appear - "pay attention to the small figures" (local) for an upper part of the screen and "pay attention to the whole figure composed from smaller figures" (global) for a lower part of the screen. Have the participant react according to the part of the screen where the stimuli appeared.
3. Add a cue indicating to which dimension (global or local) the participants should respond. Cues related to the local dimension should consist of a small red square, presented at one side of the target stimulus, and a small red rectangle, displayed at the other side of the target stimulus. Accordingly, cues related to the global dimension should consist of a big red square, presented at one side of the target stimulus, and a big red rectangle, displayed at the other side of the target stimulus.
4. Display the figures above or below the middle line in random order.
5. Have the participant react according to the previously presented rules: answering "rectangle" by using the left button and answering "square" by pressing the right one.
6. Record reaction times and correctness of the answers.

NOTE: Time for the response have to be fixed at 3500 ms. The time interval between the cue and the target stimulus should be 500 ms. The interval between the response and the presentation of the cue should be 1000 ms. Each figure and each cue should be presented for all the time needed for the participant to react by pressing one of the keys. The Switching Task measures the cognitive fluency, as it requires quick attention switching between the respective elements.

9. The linear syllogism task

1. Display on the screen a set of three 'premises' that together constitute logical chain of relations: pairs of letters with an information about a relation between them: $A > B$, $B > C$ and $C > D$. Each premise should be visible on the screen for 1500 ms and interval between them should last 3000, 3500 or 4000 ms (randomly). An integrated mental model representation³² of such set of pairs will always be in the linear order " $A > B > C > D$ ".
2. Include three pairs of possible relations between premises in separate trials: 1) $A > B$, $B > C$, $C > D$ (adjacent pairs, exactly the same as those that had been seen in the learning phase), 2) $A > C$, $B > D$ (two-step relations, not seen before and requiring integration of information), 3) $A > D$ (end point relation, not seen before and requiring integration of information).
3. Construct the task so that it contains two conditions: an easy condition, where the premises should be displayed one after the other in the order in which they form a logical string (e.g. string: $Q > W > E > R > T$, premises order: $Q > W$, $W > E$, $E > R$, $R > T$); a difficult

condition, the order of the premises should be altered (e.g. premises order: $W > E$, $Q > W$, $R > T$, $E > R$).

4. Test the participant immediately after the presentation of premises by displaying statements (for 1500 ms each) that participant needs to assess as true (answer: right button) or false (answer: left button), as fast as possible. Set time limit for the answer - 6000 ms, and after each answer given wait an additional 1000 ms before displaying the next question. Each statement should consist of only one pair of letters and relation between them ('<' or '>') in either a correct (e.g., " $W > E$?") or false setup (e.g., " $E > W$?").
5. Randomize the arrangement of the letters in order to minimize possible interference induced by implied alphabetical order of letters.
6. Use capital letters as stimuli instead of whole sentences in order to avoid linguistic connotations, and the symbol ">" to indicate the relation between elements.
7. Gather the data about the accuracy and reaction time of the response for each question.

NOTE: Questions about adjacent pairs are used to estimate memory, and questions about premises presented in a mixed order, and those asking about the relationship between the far elements of logical sequences are asked to measure the ability to information integration.

6. Training protocols

1. For both experimental (n-back) and control (Quiz) trainings provide participants with the access to the Internet platform (logins and passwords) - which allowed

them to enter the site every 24 hours, to avoid situations in which the participant trains more than once a day.

2. Make sure that participant understands the task as well as the training regimen.
3. Instruct participant to train in similar conditions every time, in calm and quiet place with possibly low level of external distractors.
4. Experimental training: working memory paradigm

NOTE: An adaptive dual n-back task served as a working memory training program. This task was introduced by Jaeggi et al.³³ and simultaneously recruits auditory and visual attention, maintenance, and updating processes.

1. Instruct participant on the task at level $N=2$ (see **Figure 1B**).
2. Use alphabet letters as auditory stimuli and green squares, presented in one of nine locations in a 3x3 matrix, as visual stimuli.
3. Present a single item for 500 ms followed by 2500 ms interval, during which the participants are supposed to respond. The current stimuli can match the target visual (response with left hand) or auditory stimulus (response with right hand) or both (response with both hands simultaneously).

4. Single session of the n-back training

1. Set level of N to 2, in the first block of the task. After each block, evaluate the correctness of the answers and, on this basis, adjust the level N in the next block. If the accuracy surpassed 85%, the difficulty level should be increased (by 1 point), if the accuracy drops below 60%, lower the difficulty level. In other cases the N remains unchanged.

2. For the first block, fix the n-back task level at $N=2$. Later, determine the N level for the current block based on the correctness of answers in the previous block. If the accuracy exceeds 85%, increase difficulty level. If the accuracy is below 60%, the level of difficulty should be lowered. In other cases, the N level should remain unchanged.

3. Set a single dual n-back session for 15 rounds (15 blocks of tasks), each with $20 + N$ trials and the whole training set for 25 sessions.

4. Record reaction times (RTs) and accuracy (ACC) measures for each trial.

5. Control training: episodic memory paradigm

1. Collect material from the Internet in order to construct the Quiz Task which engages semantic memory (e.g., what is the capital city of Hungary?).
2. Present 15 questions in each training session of Quiz Task (starting from the second session 5 questions come from the previous session and 10 should be new) with no time limit for reading it. Instruct participant that after selecting the 'answer' button they have to choose one of the four given possibilities within 40 seconds. Provide the feedback for correctness of the answers.
3. Set the whole training for 25 sessions.

7. Training supervision

1. During the course of the training verify the training progress of each participant. Assign an experimenter who is responsible for checking (online) the progress of the training, to each participant.

- If a break between sessions is longer than two days, have the experimenter contact the participant via text message and encourage him or her to resume the training.

8. Post-training assessment of cognitive functions

- Proceed with post-training session in the exact way as the pre-training meeting.
- Compensate each participant, who completed the whole protocol for the time dedicated to the research, with 150 PLN (~\$40).

Representative Results

Training-related effects

85 subjects participated in the study (29 were male) and they were on average 66.7 years old. Due to technical problems, data from one participant in the n-back training group was not recorded. Finally, the data from 43 participants in n-back training group and 42 in Quiz training group were analyzed. Multivariate analysis of variance (MANOVA) with repeated measures was used to analyze training specific effects for both training groups over time (pre-, post-training). The results of each cognitive test were dependent variables (**Table 1**), and training group and measurement points (pre-

versus post-training) were independent variables. These results are presented in **Table 2**.

The results of the analysis indicated a statistically significant post-training improvement in the syllogisms task: ($F(1,83)=31.22$, $p<0.001$, $\eta^2=0.27$) and attention switching task: ($F(1,83)=5.79$, $p=0.02$, $\eta^2=0.07$). A significant training group effect was observed for memory SPAN task ($F(1,83)=7.72$, $p=0.01$, $\eta^2=0.09$) and OSPAN task ($F(1,83)=13.01$, $p=0.01$, $\eta^2=0.14$). None of the interaction effects (time x training group) has proven to be statistically significant. However, we found significant within group effects for some analysis. In the OSPAN task, the n-back training group improved their results in second session), while for the quiz group, both performances were similar. This effect needs to be interpreted in reference to the fact that the quiz and the n-back group differed in the first measurement. Thus, results of the n-back group where initial OSPAN performance was higher improved, while the control group did not. The performance in Sternberg's and a go/no-go task was not related to a training group assignment or the time of measurement.

Overall, the results show that participants' cognitive performance was improved in the post-training execution of attention and higher cognitive functions (reasoning) engaging tests, regardless of the group affiliation.

		N-back training				Quiz training			
	session	N	Mean	Std.	Std.	N	Mean	Std.	Std.
				Err	Dev.			Err	Dev.
OSPAN	1	42	15.31	1.64	10.62	40	9.07	1.77	11.22
task	2	43	20.74	2.48	16.3	40	10	1.81	11.45

Syllogisms task	1	43	0.59	0.03	0.2	42	0.58	0.03	0.21
	2	43	0.67	0.03	0.21	42	0.69	0.03	0.19
Memory SPAN task	1	42	0.37	0.03	0.16	42	0.2	0.02	0.16
	2	41	0.4	0.03	0.18	42	0.22	0.03	0.18
Go/no-go task	1	42	0.14	0.05	0.33	42	0.16	0.03	0.01
	2	42	0.17	0.03	0.18	42	0.04	0.05	0.12
Sternberg's task	1	43	0.93	0.02	0.11	42	0.9	0.02	0.15
	2	43	0.94	0.01	0.05	42	0.93	0.01	0.07
Attention switching task	1	42	0.49	0.04	0.28	41	0.52	0.05	0.3
	2	42	0.41	0.04	0.23	42	0.46	0.04	0.25

Table 1. Descriptive statistics for the cognitive tasks' results.

	Pre- to post-training effect			Training group effect			Interaction effect (time x training group)			
	F (1,83)	η^2	p	F (1,83)	η^2	p	F (1,83)	η^2	p	significant within- group effects:
OSPAN Task	3.67	0.04	0.06	13.01*	0.14	0.01	1.49	0.19	0.22	Nback (T1 vs.T2): 5,00*
Syllogisms Task	31,22*	0.27	0	0.01	0	0.95	0.35	0.01	0.56	
Memory SPAN Task	3.13	0.04	0.08	7,72*	0.09	0.01	0.04	< .001	0.85	T1 (N- back vs. Quiz): 0,09*
										T2 (N- back vs. Quiz): 0,10*
Sternberg's Task	3.56	0.04	0.06	0.78	0.01	0.38	0.62	0.01	0.43	
Attention Switching Task	5,79*	0.07	0.02	0.75	0.01	0.39	0.02	< .001	0.87	Nback (T1 vs.T2): -0,08
Go/no- go Task	0.01	< .001	0.93	0.21	0.01	0.65	2.82	0.03	0.09	T1 (N- back vs. Quiz): -0,01

										T2 (N-back vs. Quiz): -0,02
* statistically significant effect (p < .05)										
T1 vs. T2 - difference in means between 1st and 2nd session;										
N-back vs. Quiz - difference in means between training groups;										

Table 2. Outcome measures: main and interaction effects from MANOVA with training type (n-back vs. Quiz) and time (pre vs post training) as factors.

WMC as a predictor of WM training effectiveness

In a subsequent analysis, performed on the n-back training group only, we used a more refined method - multilevel modeling (MLM) - to observe the learning process during the experimental training. The hierarchical structure of the data was accommodate to the model: at level 1 - repeated measurements, nested within participants (level 2)³⁴. The MLM dataset consisted of 1,050 observations gathered from 42 participants from experimental group within each of 25 training sessions. The model provided for both fixed and random effects: the regression intercept and slope for the average person, and between-subject variability around the average. In the Model 1, the change in the number of points scored in the n-back task over time (number of the training sessions) was modeled. The time variable was centered at 1st day of the intervention. Compared to Model 1, Model 2 added on predicting and moderating effects of a baseline OSPAN score (between-subjects predictor - level 2) on within-subject variability (level 1). Those predictors were tested independently to avoid multicollinearity. In all models,

a linear and quadratic effects for the slope were tested, however the quadratic one was subsequently removed because its fixed effects and variance components were not significant. The restricted maximum likelihood served as the estimator. -2 Restricted log likelihood ratio (-2LL) and the Akaike Information Criterion (AIC) were used to assess the goodness of fit for all models. Given the common proximal autocorrelation in the daily data³⁵ we decided to base on a first-order autoregressive [AR(1)] covariance structure.

MLM results showed that OSPAN scores from the pre-training measurement were a significant predictor of the first n-back outcome from the 1 session. Baseline OSPAN level turned out to be a moderator of the whole training course (**Table 2**). When compared, groups of participants with high or low OSPAN points had similar N-level at the first training session: approx. 2.00 units on a 1+∞ scale (low OSPAN = 1.93; high OSPAN = 2.31). A significant difference manifested in the post-training measurement, when the participants with

low initial OSPAN results achieved a .01 unit increase in n-back task, whereas those with the high initial OSPAN scores recorded a .04 point improvement. The observed result clearly indicates the existence of a positive association between the initial OSPAN level and training effectiveness. The n-back scores in the 1st session and a learning curve of a training are higher and steeper for the participants with initially higher OSPAN result ($p < .001$).

Time - linear	.031	(.005)***		.016	(.007)*
(centered at 1st day)					
Initial OSPAN score	---	---		.038	(.183)*
~ (high / low)					
Time × Initial OSPAN score	---	---		.026	(.008)**
Random effects ([co-]variances)					
Level 2 (between- person)					
Intercept	.285	(.073)***		.286	(.075)***
Time - linear	.001	(.001)***		.001	(.001)***
Intercept and time	.006	(.003)*		.004	(.002) °
Level 1 (within- person)					
Residual	.151	(.009)***		.149	(.009)***
Autocorrelation	.339	(.039)***		.328	(.040)***
Model fit					

-2 log likelihood (χ^2)	1069.32			1046.37	
Akaike's Information Criterion (AIC)	1083.32			1056.37	
Results from multilevel modeling. Unstandardized regression coefficients with standard errors in parenthesis.					
°p=0.1, *p<.05, **p<.01, ***p<.001; All p-values are two-tailed					
~ the predictor is dichotomous					

Table 3. Multilevel analysis of the training data (n-back task performance as a dependent variable). Models with training sessions (time) only (MODEL 1) and training sessions plus initial working memory capacity level as a predictor and a moderator, respectively (MODEL 2).

Discussion

In the study presented here, we have investigated whether older adults could benefit from working memory training and if it is connected to the initial level of their basic cognition. We used an n-back task as an experimental intervention and working memory capacity (measured with the OSPAN task) was the method to probe participants' initial level of intellectual functioning. We had two critical steps in the protocol. The first and most important was the assessment of the initial WM level. The second was the careful matching of the two training regimens in every possible way but the "cognitive content" (i.e., working memory versus semantic memory). By introducing the assessment of the participants' cognitive level at the beginning of the study, we were able to show how important it is to have a good estimate of it at the start of the intervention. It was the most important predictor of the cognitive training's effectiveness. We suspect that in most intervention studies, researchers assess, in one way or

another, the initial cognitive level of participants. To obtain such information it is possible to use the results from the first measurement of a trained cognitive task as a predictor of the cognitive training effectiveness. As expected, the N level of n-back task fluctuated substantially through training sessions. What was even more interesting, individuals with higher maximal N achieved in the first training session improved faster than the rest of the group in the following sessions. That implies that the variability in performance between participants, noticed at the beginning of the study, only increased with time and training. To explore this effect deeper we conducted further analysis. The results showed the preliminary score in OSPAN task (WMC) to be a strong predictor of the improvement gained during the training course (in the dual n-back task). Participants characterized by higher initial WMC performed better in the training from the very first day and had steeper learning curve in comparison to seniors with WMC below the average of the sample. We are

not the first to report such effect. Foster et al. (2017) described similar results²⁹. They proved the existence of the correlation between the initial WM level and the performance of memory span training. This result is consistent not only with the ones here, but also with research on the so-called Matthew effect in WM training interventions, in which participants with initially higher skills profit more from training and score better in both: trained and untrained, tasks^{21, 36, 37, 38, 39}. All this strengthens the conclusion that someone's ability to gain from WM training depends heavily on the initial intellectual level.

Regarding the regimens similarity, we applied the Mill's method of one difference⁴⁰: when someone observes one situation that leads to a given effect, and another that does not result in the same way, and the only difference between these situations is a presence of a specific factor only in the first situation (here, the difference in the cognitive layer), there is the solid foundation to assume that it is the factor in question that caused the observed effect. We tried to match the training regimens in terms of motivation, superficial similarities (same amount of training sessions, similar feedback, etc.). It is worth noting that the first idea was to use the same task (n-back) but in its easiest form, where the N level is fixed to 1. It quickly becomes obvious that it was a wrong path as the participants (in pilot studies) not only reported weariness but also were dropping off the control condition at a much higher rate than from the experimental (with adaptive level of difficulty). This resulted in a different approach. After several pretests we decided for a "different function" approach (WM versus semantic memory) instead of having the same function in both conditions just with different intensity (fixed level of WM versus adaptive level of WM). One potential problem with such approach is that we can create a control condition, which is more attractive than the experimental condition. And,

if motivation to engage is a crucial factor in cognitive trainings, we can have null results because of that decision.

It is worth noticing that there is a substantial change in a way we look now at results from cognitive intervention studies. For example, Reddick et al. suggest that positive effects observed in WM training groups when compared to control groups are due to decrease present in control group and not improvement of performance in experimental groups⁴¹. When we think about elderly population, even such output - maintaining of the initial cognitive level - could be a desirable outcome. But, surprisingly, in the study we did not observe a post-training reduction of performance in the control group, except for the go/no-go task. It might be, again, interpreted as evidence that even a simple memory quiz, if it is attractive and encourages participants to engage in some cognitive activity, could produce beneficial effects. What is also important, all of the participants (regardless of group assignment) volunteered for the study and some correlational studies have shown that voluntary work might be a protective factor against cognitive aging^{42, 43}. One of limitations of the study is that we do not have the representative population of older people. Instead, the elderly taking part in the study were probably more motivated and more proactive than seniors who, for example, do not leave their homes. However, the level of education and economic status (indirectly controlled - as an occupational activity that generate income) were measured in the study and the analysis showed that these were not factors affecting training progress. It can be also argued that improvement observed in both interventions is the result of mere test-retest effects. Due to the fact that there was no passive control group included in the study, this matter cannot be settled down in this study. It is therefore advisable to include another group in the subsequent tests - passive control. The most important message from the study is that the

findings suggest that post-training gains are within reach of older adults, especially those characterized by a good overall cognitive functioning. What we wanted to delineate in this article was the way we were introducing and maintaining the participants in a training regimen. The most important thing in this study was to keep all features of the intervention exactly the same between the two groups apart from one thing - the cognitive function involved undergoing practice. As we did not observe substantial differences between the effectiveness of the training protocols, but the improvement was visible in both groups, it seems valid to conclude that any cognitive engagement can be beneficial for elderly people. As the main result refers to the initial level of cognitive functioning, we strongly recommend including initial measures of the trained function and verifying it as a possible predictor (or at least co-factor) of training effectiveness.

Disclosures

The authors have nothing to disclose.

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